

Speech that Protects a Model

Typed Speech Evidence and Belief Updating in a Dialogue Dyad

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Abstract. Active-inference accounts explain how communication can support mutual prediction and belief revision, and how misunderstanding can persist. We ask a more specific question: how can an exchange restrict opportunities to reconsider a claim, and what could speech reveal about that process? We connect a typed annotation scheme, Mindloom, to a reciprocal computational dyad through an explicit observation model. Re-scoring 169 archived English cases yields macro-F1 0.757 for the typed engine and 0.837 for a direct-prompt baseline; the contribution of typing is therefore a proposed evidential discipline, not demonstrated classification superiority. Two simulated agents choose both a communicative act and a form that invites or restricts shared verification. The model separates sensitivity to delivered evidence, the cost of changing a public position, and aversion to potentially contradicting verification. Across 2,000 worlds per condition, the last mechanism reduces mean verification count from 3.646 to 2.159 and final probability assigned to the true fact from 0.885 to 0.838 despite normal verification sensitivity. High switching cost instead sustains public disagreement alongside strong private belief revision. In a 96-dialogue observer pilot, noisy form labels improve prediction of verification arrival by 0.0978 nats, but do not improve next-act prediction or mechanism classification. The access effect survives removal of information value. These results establish a computable bridge under stated assumptions, not a uniquely active-inference explanation or validation of psychological defence. We specify the human, temporal and physiological measurements needed to test the broader protective-compensatory hypothesis.

Keywords: protective-compensatory speech; dialogue annotation; active inference; belief revision; verification access; computational measurement.

1 Introduction

Friston and Frith describe communication as reciprocal inference, in which agents predict one another through a shared generative model and can modify their models through interaction [1,2]. Friston, Parr and colleagues extend this approach to linguistic exchanges, including questions and answers that provide evidence about hidden states [3]. These accounts motivate a central expectation: dialogue can help participants discover something they did not previously know. Yet continued exchange does not guarantee that a relevant claim is reconsidered. An interaction can also make further checking difficult, costly or unavailable.

We investigate this second possibility through **protective-compensatory speech**. The starting hypothesis is that protective regulation can be expressed in the organisation of speech: a participant may preserve a significant expectation partly by changing which questions, responses or evidence remain available in the interaction. The compensatory hypothesis goes further, proposing that maintaining this organisation may require continuing regulation with a physiological cost. Earlier conceptual papers in this programme develop these hypotheses [4,5]. The present paper tests a narrower computational component: whether an explicit dialogue process can distinguish reduced uptake of evidence from reduced access to evidence, and whether speech annotations provide useful observations of that process. It does not establish a physiological maintenance cost.

The distinction is necessary because public disagreement is ambiguous. A speaker who repeats a position may still believe it, may have revised a private belief while finding public revision costly, or may receive too little new information to revise it. The same ambiguity affects apparently conciliatory speech: public agreement need not imply accurate belief. An analysis that assigns psychological mechanisms directly from these surface outcomes risks treating its interpretation as an observation.

1.1 One situation and two possible trajectories

Consider a constructed exchange. It illustrates the research question and is not an observed conversation or a scored benchmark case.

Common opening. Alex: “I sent the revised budget with the latest prices.” Sam: “The attachment is version 3, with the April prices; version 4 has the May prices.”

Trajectory A — verification remains available. Alex: “Can we compare the files? I thought I had attached version 4.” Sam opens the attachment. Alex: “You are right; this is version 3. I will check the attachment next time.”

Trajectory B — verification becomes restricted. Alex: “You always look for mistakes in my work.” Alex continues: “I know what I sent. If you trusted me, you would stop questioning this.” Sam: “There is no point showing you the file.” The check is abandoned.

Trajectory A introduces a shared object of inspection. In B, questioning is made incompatible with trust, and the opportunity for inspection is lost. This is a difference in the organisation of verification as well as in whether Alex concedes. Nevertheless, B does not reveal Alex’s private belief, conscious intention or personality. Context might also justify declining a check. We therefore treat restriction as an observable candidate feature and its protective function as a hypothesis requiring additional evidence.

1.2 Related work and the specific gap

The gap is not that active inference has considered successful communication only. Medrano and Sajid’s *A Broken Duet* already models persistent misunderstanding and divergent interpretations in dyadic interaction [6]. Tison and Poirier develop communication as socially extended active inference, emphasising its ecological organisation [7]. Precision-based models also address how particular beliefs can be maintained [8]. These contributions make it necessary to specify what a new model adds beyond the general possibility of failed alignment.

Language research provides complementary tools. Dialogue-act annotation distinguishes communicative functions [9], and conversation analysis treats contributions in relation to the sequences in which they occur [10]. Probabilistic pragmatics models how listeners infer meaning from a speaker’s choices [11]; models of politeness show how competing social goals can shape an utterance [12]. Work on defensive communication places defensiveness within interaction [13]. More recently, Na and colleagues introduced PsyDefConv and an annotation pipeline for levels of psychological defence in supportive conversations [14]. That task is close in subject matter but uses different labels, data and validation criteria; its scores are not comparable with those reported here.

Our focus is the connection between **an annotated interactional form, its consequences for access to evidence, and competing explanations of public persistence**. A label alone does not specify the process that generated it. A formal agent model alone does not specify how its hidden variables could be observed in language. We connect these levels by making invitation and restriction actions within a reciprocal process, then treating annotations as noisy measurements of those actions. This gives an explicit answer to three questions: what is observed, what is inferred, and what is assumed.

1.3 Contributions and scope

Part I defines typed speech observations and reports an archival evaluation with its provenance limits. Part II specifies a dyad in which a partner’s move provides evidence and jointly chosen forms influence subsequent verification. An observation model connects the two parts without assigning labels directly to hidden psychological parameters. Computational experiments then examine both the effects of the stipulated mechanisms and the incremental predictive value of the labels.

The central result is a separation between **private factual revision, public expression and opportunities for verification**. All mechanistic results concern synthetic agents. “Belief updating” refers here to Bayesian revision of a binary factual belief and inference over a fixed partner dictionary. The model does

not learn the structure of its likelihoods or demonstrate long-term revision of a person’s generative model. The broader language of model protection names the research hypothesis, not an already validated human construct.

2 Part I Describing speech with explicit evidence

2.1 Annotation units and evidence requirements

Mindloom annotates a speaker’s contribution in the context supplied to the system. Its proposed discipline is to make the path from evidence to output inspectable. The pipeline records surface evidence atoms at L1, aggregates features at L2, checks label-specific evidence requirements at L3, assigns at most one configuration at L4, and emits a typed result with an evidence trail, or abstains, at L5. A rule that requires evidence for a label is called a *bridge* in the engine; it is distinct from the probabilistic observation bridge introduced in Section 4.

The ontology contains eight configurations, a closure function and a transition event (Table 1). Keeping these types separate matters. A question form is not itself evidence of a transition, and ending a topic is not necessarily the same action as constraining an interlocutor’s acceptable replies. An annotation can describe such a constraint without establishing why the speaker produced it.

Table 1. Annotation targets and their evidential scope. These are descriptive targets, not diagnoses or inferred motives.

Type	Label	Descriptive target
Configuration	BUILD	Joint elaboration or repair while a response remains available
Configuration	SEEK	Inquiry that makes an answer relevant
Configuration	UNSEAL	First-person disclosure without requiring the other to regulate it
Configuration	LOCK	Language constraining the addressee’s acceptable replies
Configuration	DRAIN	The other positioned as a required source of support or regulation
Configuration	FLOOD	Intensification or repetition dominating the exchange
Configuration	EDGE	Alternation between approach and withdrawal
Configuration	VOID	Withdrawal from an offered opportunity to respond
Function	SEAL	Closure of a topic or channel
Event	SHIFT	Change supported by before-and-after configuration evidence

SEAL can co-occur with a configuration. SHIFT requires temporal evidence; its appearance as a primary label in a historical classification task does not turn it into a configuration. *Insufficient context* is an abstention output, not an eleventh psychological category. The term *regime* occurred in earlier versions of the ontology, but persistent dynamical regimes have not been established by this evaluation.

The later simulation uses only two provisional anchors: SEEK for an invitation to verification, and LOCK for restriction. This is a deliberate reduction of scope. SEEK in ordinary conversation is broader than a request to verify a disputed fact, and LOCK is broader than the particular restriction implemented below. Whether these anchors transfer to the model’s two forms remains an empirical question.

2.2 Evaluation material and provenance

The supplied benchmark documentation describes 589 English monologue fragments generated for the ontology and fixed before the reported runs. They are not recordings of natural conversations. Tier A comprises 169 cases used for primary-label evaluation; Tier B comprises 420 cases used for defence-recall testing. These benchmark tiers are separate from the evidence layers L1–L5. The present classification analysis and observation-channel calibration use Tier A only.

The supplied construction summary reports word-count minimum/median/maximum of 35/50/71 for Tier A and 1/15/47 for Tier B. It reports no exact duplicates, no Tier A near-duplicate pairs, and 15 Tier B near-duplicate pairs under a word 5-gram Jaccard threshold of 0.3. These are documented corpus descriptors, not a new corpus audit in this revision. A count of pairs does not establish the proportion of unique cases involved.

Fixing cases before a run is weaker than demonstrating independence from ontology development. The available records do not fully establish who generated and labelled each case, independent adjudication, or separation between ontology designers and gold-label authors. The appropriate interpretation is an internal curated evaluation. It cannot establish performance on naturally occurring dialogue or an independently sampled population.

2.3 Archived classification results

We use the July 2026 stored predictions for a typed engine and a direct-prompt baseline. There are three predictions per case. Label majority determines the case output, with run 0 breaking a three-way tie. Strict scoring accepts the designated gold primary label only. Macro-F1 averages the ten gold categories; an abstention counts against the relevant gold category. No new language-model calls are used for this analysis.

Table 2. Strict primary-label classification on 169 archived Tier A cases. Fractions count correctly classified cases. Macro-F1 uses all ten gold categories.

System	All cases	Non-SHIFT cases	Macro-F1	SHIFT cases
Direct prompt, majority of three	145/169	127/149	0.837	18/20
Typed engine, majority of three	129/169	127/149	0.757	2/20
Majority gold class, BUILD	20/169	20/149	0.021	0/20

The direct prompt performs better overall; both systems correctly classify 127 of 149 non-SHIFT cases. This equality concerns accuracy, not necessarily the same case-level predictions or per-class performance. The typed system has no demonstrated classification advantage. Its proposed benefit is traceable evidence and control over permissible output types, which requires a separate evaluation of evidence quality and unsupported inference.

SHIFT exposes a particular weakness. The typed majority output is SEEK on 17 of the 20 transition cases. Run 0 abstains on 15 of those cases, whereas runs 1 and 2 do not abstain. Majority voting can therefore conceal variation in abstention behaviour. Future evaluation should report coverage and error conditional on coverage, together with transition-specific performance.

An earlier manuscript also reported 43/169 correct cases and macro-F1 0.195 for keyword rules revised three times on this same set. This is a test-tuned exploratory comparator, not a held-out estimate or an upper bound on lexical methods. Its executable rule set is not part of the present reproducibility package, so it is not included in Table 2. The result cannot establish that the labels are unrecoverable from surface features.

2.4 Reliability and architectural claims

A six-rater pilot in the supplied internal report evaluated evidence features rather than the ten labels. Reported Fleiss' kappa was 1.00 for lexically direct features such as a named feeling and an explicit return time, 0.23 for minimal response, 0.13 for absolute expressions, and -0.11 for required response. Raw rater matrices are unavailable in the supplied package, so these values are transcribed rather than recomputed. They indicate that the more relational features need clearer instructions and stronger validation. They do not provide a calibrated noise model for the ten output labels.

In a separate auxiliary output family, an archived gates-off capture contains 17 of 65 manipulation-label emissions whose cited bridge lacked a required item in the recorded L1 evidence. These auxiliary

annotations are distinct from the primary labels in Table 1. The statement that the gated version removed all 17 is author-reported and cannot be recomputed as a matched comparison from the supplied material. A controlled ablation using the same model, prompt conditions and test set is still needed. We therefore present typing as an architectural proposal whose effect on unsupported inferences remains to be measured, not as an established causal explanation of the classification scores.

3 Part II A reciprocal model of updating and verification

3.1 Why public persistence leaves the mechanism open

The simplest intuition does not require a dyad. If a speaker assigns high probability to a claim, it may repeat it because the claim appears accurate. If its private belief changes, an additional cost of publicly reversing position may still favour repetition. A third possibility arises once actions influence observation: a speaker can retain normal sensitivity to evidence while selecting interactions that produce fewer opportunities to check the claim. These possibilities concern evidence uptake, expression and evidence access, respectively.

The reciprocal model represents all three in one process. It does not assume that they produce identical behaviour, and its parameter settings were not fitted to match a single non-concession rate. Public outcomes are compared alongside private beliefs and verification events. This avoids inferring the mechanism from the very label used to define it.

3.2 World, private records and event order

There is a fixed physical fact $x \in \{0, 1\}$, sampled with probability one half. A initially assigns probability 0.75 to $x=1$, and B assigns probability 0.25. Each receives $N=2$ private binary records, independent conditional on x , with reliability $\rho_p=0.75$. Let s_i count agent i 's records equal to 1. Both incorporate these initial records at the same reliability. Thus any later attenuation effect does not arise from different treatment of the initial data.

Eight rounds contain sixteen alternating moves, A then B. At each move, the speaker chooses an act and a form. The listener observes both and updates its beliefs about the fact and the partner. The speaker's public position is updated. A fresh verification outcome may then arrive; if it does, both agents receive the same outcome once. Roles alternate. Initial public positions are A:1 and B:0; the preceding move for each agent is initialised as inviting SILENCE.

A move is $y_t = (u_t, f_t)$. Its act is one of STATE_0, STATE_1, ASK, SILENCE or REASSURE. Its form is restrictive ($f_t=0$) or inviting ($f_t=1$), giving ten possible moves. These are idealised actions, not generated sentences. A concession is a factual statement that changes the speaker's **current** public position. It is not a permanent action name tied to the initial claim.

Let f_j^- and u_j^- denote the partner's most recent form and act. The arrival probability of shared verification after a move is

$$\lambda_t = \lambda_{\min} + (\lambda_{base,t} - \lambda_{\min}) f_t f_j^-, \lambda_{\min} = 0.05. \quad (1)$$

Here $\lambda_{base,t} = 0.90$ if the current or preceding partner act is ASK, and 0.35 otherwise. Restriction reduces arrival to a non-zero floor; verification does not require both invitations with absolute certainty. The dependence of arrival on form is stipulated. It is the main causal hypothesis that would have to be estimated in real interaction.

3.3 Three parameters with different roles

Each participant has its own $\theta_i = (\omega_i, c_i, \gamma_i)$. The four focal parameter points are shown in Table 3. They are experimental settings, not psychological types of people.

Table 3. Focal profiles in the computational experiments. Each parameter may also differ between the two participants.

Profile	ω	c	γ	Changed component
Reference	1.0	0	0	None of the three additional mechanisms
Verification attenuation	0.3	0	0	Subjective reliability of new verification
Public switching cost	1.0	1.4	0	Cost of changing the current public position
Access protection	1.0	0	2.0	Cost of potentially contradicting verification

For actual agents, ω_i adjusts the subjective reliability of a newly delivered verification outcome while preserving its arrival probability:

$$\rho_{\omega_i} = \sigma(\omega_i \logit \rho_v), \rho_v = 0.75, \quad (2)$$

where σ is the logistic function. Conditional on the chosen moves, the subjective likelihood for $e_t \in \{0, 1, \emptyset\}$ is

$$A_i(e_t \mid x, y_t, y_j^-) = \begin{cases} \lambda_t \rho_{\omega_i}, & e_t = x, \\ \lambda_t (1 - \rho_{\omega_i}), & e_t = 1 - x, \\ 1 - \lambda_t, & e_t = \emptyset. \end{cases} \quad (3)$$

The physical generator uses ρ_v rather than ρ_{ω_i} . Equation (2) attenuates the evidential strength of a signal, without changing the predicted chance that any signal arrives. Initial private evidence is unchanged. Actual interpretation of partner speech uses the joint Bayesian model below, not a manually adjusted marginal posterior. Accordingly, ω should not be read as a measured general sensitivity to all information.

The parameter c_i penalises changing the current public position. A preceding REASSURE by the partner sets this cost to zero for the recipient's next choice only. The parameter γ_i prices the expected occurrence of verification contradicting the position that would hold **after** the proposed move. Neither parameter represents metabolic expenditure, an inferred motive or measured allostatic load.

3.4 Beliefs about the fact and the partner

Each agent maintains a joint posterior

$$J_i(x, s_j, \theta_j), J_{i,0} = P_i(x \mid s_i) P(s_j \mid x) P(\theta_j). \quad (4)$$

The internal partner dictionary contains all eight combinations of $\omega_j \in \{0.3, 1\}$, $c_j \in \{0, 1.4\}$ and $\gamma_j \in \{0, 2\}$, with a uniform initial prior. The agent's factual belief is $q_i = P_i(x=1)$ obtained by marginalising J_i .

Both actual agents use a bounded level-1 model: each infers a partner represented as a level-0 chooser driven by expected costs. Their partner models are deliberately approximate because the actual partner also values information in the main condition. This is not an equilibrium of recursively correct models of one another.

The predicted partner policy depends on the hypothesised private-record count, the partner parameters, public commitments, shared verification already received and factual messages already sent. Details appear in Appendix A. After observing a partner move y_j , the listener updates

$$J_i^+(x, s_j, \theta_j) \propto J_i(x, s_j, \theta_j) \pi_j^{(0)}(y_j \mid s_j, \theta_j, h_t). \quad (5)$$

Here h_t denotes the available public history. After a delivered verification outcome, both agents multiply their respective joint distributions by (3) and normalise. Once chosen moves are known, the no-signal likelihood is constant in x and cancels in normalisation.

This construction makes the partner's speech evidence without treating each repeated assertion as a fresh independent private record. Speech informs a fixed, finite latent record and partner profile. It can improve inference, but the information available from those records is bounded; an independent check tests that repeated assertions cannot create unlimited private evidence (Appendix A).

3.5 Expected costs and epistemic value

Let $v_i(y)$ be the public position after candidate move y : the stated value for a factual act, otherwise the previous position. The ordinary expected loss is

$$L_i(y) = \begin{cases} k P_i(x \neq v_i(y)) + c_i^{eff} 1[v_i(y) \neq v_i^-], & u \text{ factual,} \\ b, & u \text{ non-factual,} \end{cases} \quad (6)$$

with $k=2$, $b=1.1$ and c_i^{eff} set by the current reassurance state. The full expected cost is

$$R_i(y) = L_i(y) + \delta(1-f) + \gamma_i \lambda(y) P_i(e \neq v_i(y) \mid e \neq \emptyset, y), \delta=0.12. \quad (7)$$

The contradiction probability uses the same subjective reliability as the update. The restriction cost prevents restrictive form from being costless. Because the arrival floor is positive, the contradiction term can remain positive even for a restrictive move.

The planning window includes the verification outcome E immediately after the candidate move and the partner's subsequent response Y_j . It ends before verification following that response. For latent state $\ell = (x, s_j, \theta_j)$,

$$p_i(E, Y_j \mid \ell, y) = A_i(E \mid x, y, y_j^-) \pi_j^{(0)}(Y_j \mid s_j, \theta_j, h_t, y, E). \quad (8)$$

These probabilities define a normalised distribution over thirty possible outcome pairs. The information term is mutual information under this same model:

$$I G_i(y) = I_{J_i}(\ell; (E, Y_j) \mid y), G_i(y) = R_i(y) - \beta I G_i(y), \quad (9)$$

$$\pi_i(y) = \frac{\exp[-G_i(y)/\tau]}{\sum_{y'} \exp[-G_i(y')/\tau]}, \tau=0.20. \quad (10)$$

The main condition uses $\beta=1$. A cost-only control sets $\beta=0$; another control values information about verification alone. Joint information gain concerns the fact, the partner's private record and its profile, so maximising it need not maximise factual accuracy alone.

Equations (8)–(10) instantiate an extrinsic-minus-epistemic choice objective used in discrete active inference [15]. They specify a likelihood, a posterior update and an information-based policy. The model fixes policy temperature and does not implement the full neuronal process theory or infer policy precision [16]. Its choice rule is also mathematically expressible as expected utility plus information value. Formalisation therefore addresses whether the link is computationally explicit; comparison with behaviourally distinct alternatives is needed to establish explanatory advantage.

4 Connecting annotation to the dyadic process

4.1 The observation model

The model produces idealised acts and forms. To connect it to Part I, we add a noisy label z_t emitted from the current form:

$$z_t \sim \text{Categorical}(C_{f,:}), C_{f,:} \sim \text{Dirichlet}(n_{f,:} + \alpha_0), \alpha_0=0.5. \quad (11)$$

The restrictive row uses the historical gold-LOCK counts: 10 LOCK, 3 SEAL and 2 BUILD, from 15 cases. The inviting row uses the gold-SEEK counts: 20 SEEK from 20 cases. Each row has eleven output cells, including abstention, and receives 0.5 in each cell. Thus the perfect empirical SEEK row does not

imply known perfect measurement. This is uncertainty about finite calibration counts, not a fixed arbitrary mixture of informative and uninformative labels.

One matrix C is drawn for each posterior-predictive dialogue and held fixed across all of its moves and both speakers. The external observer integrates uncertainty in this matrix; it is not given the sampled matrix. This construction averages over plausible calibration matrices. It does not claim that a deployed engine changes its confusion probabilities independently between conversations.

The annotation is conditionally independent of θ given the form. A label is not generated directly from a hidden mechanism, private belief or allostatic state. Any information about these parameters must arise through their consequences for action and form. Conversely, because form is stipulated to affect verification, the construction already makes form potentially relevant to predicting arrival. The simulation evaluates transmission through a specified noisy channel, not the existence of that relation in people.

The eleven possible output names are classification outcomes, including mistakes. An output named SHIFT does not mean that the simulator generated a latent SHIFT event. There is no full eight-regime latent process, natural-language verbaliser or re-annotation of simulated text. The bridge therefore tests two idealised forms rather than validating the entire ontology.

4.2 What the observer knows

Table 4. Observation and inference boundaries. All three external-observer conditions receive coarse acts and past verification outcomes.

Component	Status in the external analysis
Acts, speaker order and delivered verification	Observed in every channel
Current annotation	Observed only in the label channel, after the act and before verification
True interactional forms	Observed only in the form-oracle condition; otherwise latent
Private record counts and physical fact	Latent and marginalised; not exposed to the observer
Parameters of A and B	Both unknown; inferred on a specified finite dictionary
Choice model, event order and form-to-access rule	Assumed known
LOCK/SEEK anchoring and calibration transport	Assumed; uncertainty in finite row counts is integrated
Motives, clinical defences and physiological load	Neither measured nor recovered

The external parameter dictionary contains the four distinct points in Table 3, with a uniform prior for each participant. Crossing sixteen parameter pairs with nine private-count pairs gives 144 static strata. The shared physical fact induces dependence between the two private counts and is marginalised accordingly. The observer uses the actual level-1 choice model and the physical verification reliability. Its dictionary is distinct from the eight-point dictionary used *inside* the agents’ approximate partner models.

We compare **acts plus evidence**, **acts plus evidence plus labels**, and **acts plus evidence plus true forms**. Calling the first condition “actions only” would be inaccurate: past verification is available in every condition. The oracle is a within-model measurement reference, not an empirically attainable performance estimate.

4.3 Filtering and prospective predictions

With observed forms, inference is exact on the finite grid. With hidden forms, histories branch over possible forms and require approximation. The main run maintains $K = 8$ path particles within each of the 144 parameter/private-count strata. Static strata are retained, while form histories are resampled within

them. A fully adapted proposal uses the current act, label if available, and verification outcome. Dirichlet rows are analytically integrated through form-specific label counts along each path. This is a stratified particle filter, not an exact hidden-form filter.

Prediction is separated from the subsequent posterior update. The next-act forecast uses only the prefix ending before that act. The verification-arrival forecast additionally uses the current act and, where supplied, its current annotation, but is scored before the new verification outcome. No future labels, later evidence or private participant beliefs enter these forecasts.

Two null controls clarify the claim. If the label distribution is identical across forms, the label carries no additional information about form or mechanism. If form no longer affects verification in this specification, and the remaining form cost is a constant additive term, form becomes uninformative about the parameters conditional on the act. These are mathematical checks of the implemented channel, not tests that natural-language annotations satisfy the assumed mapping.

5 Computational evaluation

5.1 Design and outcome measures

All results in Sections 5 and 6 use the **Hybrid reciprocal verification dyad v0.8, dated 30 September 2026**, specified above. They are not pooled with earlier single-agent or dyadic versions. The protocol is an exploratory development record, not an externally registered preregistration.

Dynamics experiments use 2,000 shared random worlds per condition. The 49 main conditions include all sixteen pairings of Table 3 under $\beta=0$ and $\beta=1$, three verification-only-information conditions, twelve targeted intervention conditions, and two no-gating controls. Four additional conditions vary verification reliability to 0.65 or 0.85 for reference and access-protection pairs. Shared random numbers support paired comparisons; worlds are independent within the simulated population. These are simulated worlds, not independent human dyads.

The principal factual measure is final probability assigned to the true state, averaged over both agents and worlds. We also count delivered shared verifications, measure final absolute belief difference $|q_A - q_B|$, and record whether final public positions disagree. Public disagreement is not equivalent to private disagreement or to an unchanged position throughout the exchange. All uncertainty intervals reported below are paired Monte Carlo intervals, calculated as the mean world-level contrast plus or minus 1.96 standard errors. They describe finite simulation sampling, not population uncertainty about humans or a multiplicity-adjusted confirmatory analysis.

The external-observer pilot uses a separate seed and 96 dialogues, balanced across the four generating A profiles. B is generated from the reference profile but is **unknown to the observer**, which infers both profiles. Consequently, this pilot does not evaluate all sixteen generating pairings. Forecast losses average moves 5–16 within each dialogue and then across dialogues. Profile classification uses the maximum posterior marginal after all sixteen moves.

5.2 Private revision and public expression diverge

Table 5. Homogeneous pairs under the joint-information policy. Means over 2,000 worlds; both agents have the listed profile. The common initial mean probability assigned to truth is 0.679.

Profile	Final truth probability	Verifications	Final belief gap	Public disagreement
Reference	0.885	3.646	0.099	8.7%
Verification attenuation	0.819	3.576	0.163	12.5%
Public switching cost	0.907	4.911	0.067	78.9%
Access protection	0.838	2.159	0.136	11.8%

The high-cost pair displays the largest public disagreement and the smallest private belief gap. Its final truth probability exceeds that of the reference pair by 0.0213 (95% interval 0.0147 to 0.0279). Thus persistent public disagreement does not establish failed private revision. The cost changes how revised beliefs are expressed and also alters which actions are selected; it is not a pure change in a reporting channel.

Attenuation and access protection both reduce final factual performance, but through different model components. Attenuation weakens the use of new verification. Access protection leaves verification sensitivity at $\omega=1$ and changes its supply: the pair receives 1.487 fewer verification outcomes than the reference pair (interval -1.546 to -1.428) and assigns 0.0470 less probability to truth (-0.0540 to -0.0400). It still improves substantially from the initial 0.679. The result is reduced updating opportunity and reduced final learning in this task, not the complete absence of learning.

The access-protection and reference pairs are not publicly identical: their disagreement rates differ by 3.1 percentage points (interval 1.7 to 4.5). The relevant conclusion is that one public summary does not uniquely identify the internal process, not that these generators are observationally equivalent.

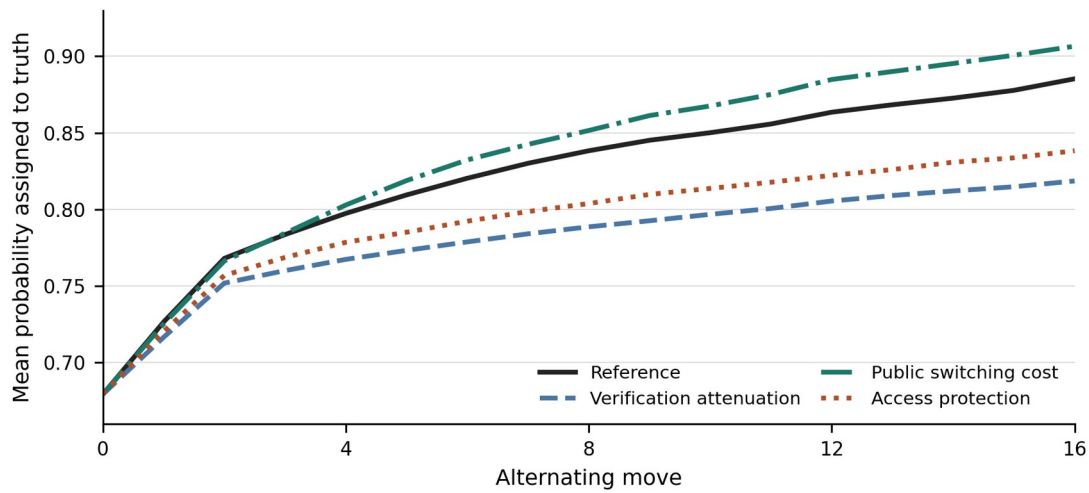


Figure 1. Belief revision across the sixteen alternating moves. Curves are means over the same 2,000 simulated worlds. Public switching cost can coexist with strong factual revision; access protection reduces revision while retaining normal sensitivity to delivered verification.

5.3 Controls and targeted changes

Removing information value does not remove the access effect. At $\beta=0$, truth probability is 0.867 for reference pairs and 0.831 for access-protection pairs; mean verification counts are 3.070 and 1.850. The paired truth difference remains -0.0356 (-0.0419 to -0.0292). The epistemic term changes the dynamics, but it is not necessary for the core reduction in verification. The current results therefore cannot establish that active inference explains this pattern uniquely.

Forcing inviting forms from the eighth move, while preserving the selected coarse acts, raises access-protection truth probability to 0.891 and mean verification count to 4.198. This intervention directly changes a variable that controls access in Equation (1). It demonstrates the causal role of that variable *within the simulator*; it is not evidence that instructing people to use a particular phrase produces the same effect.

For a high-cost A interacting with a reference B, forcing one effective REASSURE at B's fourth move reduces final public disagreement to 18.75%, compared with 44.85% for a local inactive-message control. Both conditions have the same earlier history under shared randomness; the inactive message differs only in whether that move removes the recipient's next-choice switching cost. It is a known inactive control, not a deceptive psychological placebo. Truth probability is 0.886 with effective reassurance and 0.888 with the inactive control, so a public change should not be described as improved factual learning.

The access-protection reduction in verification and final truth probability also occurs at sensor reliabilities 0.65 and 0.85. These two checks establish local robustness to that parameter, not robustness across every plausible task or model. The complete mixed-pair and intervention tables are supplied with the model results; they should not be read as a human typology or a phase diagram of clinical states.

5.4 What noisy speech labels add

Table 6. External-observer pilot on 96 independent simulated dialogues. Every channel observes coarse acts and past verification. Lower log loss is better; losses are in nats. A-profile accuracy concerns four focal points.

Additional observation	Arrival log loss	Next-act log loss	A-profile accuracy
None	0.4991	0.4295	47.9%
Noisy form labels	0.4013	0.4307	46.9%
True forms	0.3639	0.4302	49.0%

Labels reduce verification-arrival log loss by 0.0978 nats (95% paired interval 0.0743 to 0.1213), approximately 19.6% relative to the acts-plus-evidence condition. The true-form gain is 0.1352 nats (0.1081 to 0.1623). Under the stipulated access rule, the noisy channel preserves useful information about an upcoming environmental event.

The result does not extend to every inference target. The next-act gain is -0.0012 nats (-0.0039 to 0.0015); the A-profile accuracy difference is -1.04 percentage points (-9.99 to 7.90). Mean posterior probability assigned to the true A profile increases by 0.0196, with an interval including zero (-0.0091 to 0.0483). Thus this pilot supports improved prediction of verification arrival, not improved mechanism identification or next-act prediction.

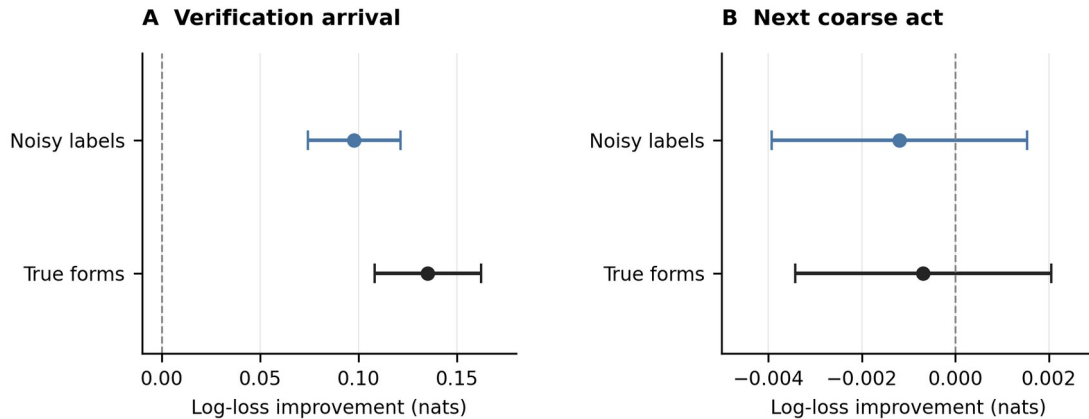


Figure 2. Incremental forecast value of labels and true forms over acts plus evidence. Positive values indicate lower log loss. Intervals are paired 95% Monte Carlo intervals over 96 dialogues. Arrival and next-act forecasts use different prediction times as specified in Section 4.3.

This distinction is central to the article’s measurement claim. An annotation can help describe how an exchange controls access to evidence without revealing the hidden reason for that organisation. Conversely, identifying a generator within a small known dictionary would not establish that the same label diagnoses a corresponding human mechanism.

5.5 Numerical validation and approximation limits

Thirteen independent checks pass in the supplied validation record. Scalar and vector calculations agree to 2.7×10^{-15} for choice probabilities and 2.0×10^{-15} for mutual information. Other checks cover Bayesian averaging, likelihood normalisation, role symmetry with turn-order reversal, bounded information from private records, the causal role of current public position, local-control timing, Dirichlet integration, and null channels.

Short exhaustive enumeration over 576 final hidden-form paths differs from the particle filter by at most 0.0023 in posterior components and 0.0029 in the tested arrival forecasts. This validates a small example, not exactness on full dialogues. Increasing paths from 8 to 32 on sixteen dialogues yields an arrival-label gain of 0.0832 nats, compared with 0.0771 at 8 paths. Individual posterior components change by up to 0.124 in the acts-plus-evidence channel and 0.060 in the label channel. Profile classifications therefore remain preliminary, and full-sample convergence has not been established. Appendix C gives the checks and sensitivity scope.

6 Discussion

6.1 The substantive link between the two parts

The two parts address different sides of one measurement problem. Part I asks whether an analyst can identify a contribution’s evidentially supported interactional form. Part II asks how such a form could matter for belief revision and what other observations are needed to explain it. Equation (11) connects the annotation output to a specified hidden action; Equation (1) connects that action to an opportunity for verification. The connection is explicit enough to simulate, remove and test.

The most useful model result is the access route. An agent need not lose the capacity to update in order to end with less information. When anticipated contradiction influences action choice, the interaction can reduce the availability of the very observations that would support revision. This gives a precise candidate meaning to the original idea that protection can appear in speech. Its empirical content is a conditional prediction about verification and belief change, not a claim that a word or label reveals a defence mechanism.

The public-cost result supplies the complementary warning. A participant can retain a public position while privately revising a belief. “Not conceding,” “not checking,” and “not learning” must therefore be measured separately. In human exchanges, any of these may also reflect justified distrust, unequal authority, limited time or other situational demands. The present model cannot adjudicate these explanations without additional observations and alternatives.

6.2 What active inference contributes and what remains unproven

Active inference motivates representing communication as action that changes future observations and selecting actions through both expected consequences and information value. Here this motivation is implemented by a normalised generative likelihood, Bayesian updates and an explicit epistemic objective. This is stronger than assigning active-inference terminology to speech categories after classification.

However, the cost-only control shows that the main access-protection effect does not depend on epistemic value. Moreover, an expected-utility model with the same information bonus has the same policy. No behavioural experiment can distinguish two formulations that make identical predictions. Evidence for the epistemic component would instead require comparison with alternatives that differ in predictions, under conditions that vary uncertainty, information availability or planning horizon while controlling expected external costs. Parameter recovery, model recovery and held-out prediction should precede psychological interpretation [17].

A useful future test would ask whether the joint-information model predicts responses to *informative but non-rewarding* dialogue opportunities better than cost-only and simpler uncertainty-sensitive policies. The tests must also allow the active-inference specification to fail. Better fit to data generated by the same model, or persistence of a result after an ablation, is not evidence of theoretical uniqueness.

6.3 Why the compensatory hypothesis remains a roadmap

The broader proposal links sustained protection to compensatory regulation. Work on uncertainty and stress motivates considering continuing regulatory demands [18], while allostatic-load research concerns cumulative multisystem biological burden [19]. Neither literature supplies the missing measurement in

this study. Our costs are dimensionless preference terms, not energetic expenditures; no physiological state, recovery process or accumulation of biological burden is simulated or observed.

The model therefore supports only one component of the programme: interaction can preserve reduced exposure to revising evidence under a specified policy. Whether maintaining such interaction is costly to an organism, whether costs accumulate, and whether any effect is specific to protection rather than continued external demand remain open. Adding a variable called “allostatic load” without independent measurement would not resolve these questions.

6.4 Limits of interpretation and transport

The dyad concerns a static binary fact, a short exchange, a small action set and fixed parameters. Real dialogue includes changing topics, ambiguity, asymmetrical authority, deception, memory limitations and learning of the model itself. Both actual participants use intentionally misspecified partner models. The external observer, in contrast, is given the correct process family and a small candidate dictionary; this makes its task easier than open-ended inference from human conversation.

The historical LOCK and SEEK rows are primary-label errors on curated fragments, not independent calibration of invitation and restriction in recorded dyads. Dirichlet uncertainty represents finite counts only. It does not capture corpus shift, coder disagreement, dependence of errors on context, or misspecification of the ontology. There is no end-to-end test from real speech through the engine to a validated latent mechanism.

Finally, the benchmark’s weak SHIFT performance and the lack of incremental next-act prediction constrain claims about trajectories. Sequences of labels are observable sequences; calling them dynamical regimes requires evidence of persistence, recurrent transitions and predictive value beyond recent context. No such empirical demonstration is offered here.

7 Roadmap for empirical tests

7.1 Validate the observation before inferring the mechanism

The first study should use a narrowly defined factual task with genuinely checkable evidence. Independent coders should annotate whether each contribution invites, restricts or leaves unchanged a specific verification opportunity, using only the dialogue prefix available at that time. Definitions should allow uncertainty and abstention. Expert adjudication, coder training and held-out assessment should be separated, and the natural-language engine should be compared with both direct prompting and simple dialogue-act features.

The crucial observation-model test is whether these annotations predict an independently recorded opportunity for checking, conditional on the act, task and available evidence. The relation should be estimated rather than fixed as Equation (1). The LOCK/SEEK anchoring can then be retained, revised or rejected. A code should not be accepted merely because it correlates with a later label derived from the same ontology.

7.2 Separate uptake, expression and access experimentally

A factual task can provide participants with distinct initial records, elicit private probabilities separately from public statements, and record which verification opportunities were offered, requested, received and inspected. Evidence reliability or availability and the cost of changing a public commitment should be varied independently where feasible. This yields observations capable of separating three hypotheses: weak belief change after equivalent delivered evidence; belief change without corresponding public revision; and reduced evidence receipt despite preserved sensitivity when evidence is supplied.

Private belief elicitation is itself a measurement with possible reactivity and reporting noise; it should not be treated as a perfect window into belief. The study needs a measurement model and manipulation checks. Preregistration should follow task piloting and parameter/model recovery under plausible effect

sizes, missingness and measurement errors. The simulated numbers in this paper do not justify a human sample-size recommendation or a claim that a mechanism can be diagnosed from a small number of conversations.

7.3 Test trajectories and compare models prospectively

Longer repeated interactions are needed to test whether configurations organise into stable or recurrent regimes. Speaker-level and dyad-level units should be specified in advance. Candidate trajectory models should predict inquiry, repair, closure, concession or verification from a strict prefix and be evaluated on held-out participants and dyads. Baselines should include the last act, a recent-context model, and a model with the same acts and evidence but without the proposed regime variable.

Model comparison should include cost-only policies, alternative learning rules, bounded planning and richer partner dictionaries. Mixed mechanisms and changing parameters should be included in recovery studies rather than forcing every participant into one focal profile. A transition annotation and a change in a latent dynamical regime should remain distinct test targets. Failure to improve held-out prediction would count against adding the regime variable.

7.4 Measure acute regulation and cumulative burden separately

An initial physiology study could align dialogue events with cardiac, respiratory and electrodermal recording, perceived effort, and baseline and recovery periods. Analyses would need to account for speaking duration, movement and continuing external demands. A testable possibility is that restricting verification reduces immediate discomfort while prolonging subsequent regulation; faster recovery or no difference must remain admissible outcomes.

Allostatic load requires a separate longitudinal design with repeated exposure assessment, relevant covariates and a prespecified multisystem biological index [19]. Acute arousal during one difficult exchange is insufficient. Only after an independent observation model is supported should physiological variables be linked to repeated speech trajectories and their putative maintenance costs.

8 Conclusion

Protective-compensatory speech poses a measurement problem before it poses a classification problem. Public persistence alone does not reveal whether a belief failed to change, changed without public expression, or received too little new evidence. We connect typed speech evidence to an explicit reciprocal process in which interactional form controls opportunities for shared verification. Within that model, access protection reduces factual revision despite normal sensitivity to delivered evidence, and noisy labels improve prediction of verification arrival. Mechanism recovery, next-act prediction, human construct validity and allostatic cost remain unestablished. The contribution is a formal and falsifiable connection between annotation and interaction, together with the measurements needed to test its wider psychological interpretation.

Data and code availability

The accompanying reproducibility package contains the v0.8 specification, protocol, simulator, external observer, independent checks, completed result tables, figure scripts and the manuscript source. It also includes the archived engine and direct-prompt predictions used for re-scoring, the historical confusion counts and their provenance. Exact reruns of those language-model calls are not possible from the available metadata: sampling settings and complete provider/version records were not retained. Re-scoring the stored outputs requires no model calls.

The dynamics and observer results reported here belong to the single version named in Section 5.1. Older single-agent and reciprocal-dyad results are not used as quantitative support for this specification. The internal pilot reliability values and corpus-construction descriptors have the limited verification status stated in Section 2. The project repository is

https://github.com/muqali2816/mindloom_IWAI_2026_speech_protects_model; the accompanying package identifies the exact files underlying this manuscript, without presuming that a matching public release has already been deposited.

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Appendix A Formal details

A1 Initial beliefs and the level 0 partner model

Let $a_A = \text{logit}(0.75)$ and $a_B = -a_A$. Actual initial factual beliefs are

$$\text{logit } q_{i,0} = a_i + (2s_i - N) \text{logit } \rho_p. \quad (\text{A1})$$

For a candidate partner record s_j and profile θ_j , the approximate level-0 partner belief uses

$$\text{logit } \tilde{q}_j = a_j + (2s_j - N) \text{logit } \rho_p + m_E \text{logit } \rho_{\omega_j} + m_i \omega_j \text{logit}(0.70). \quad (\text{A2})$$

Here m_E is the signed count of shared verification outcomes and m_i is the signed count of factual messages already sent by i . The approximation treats messages received by the modelled level-0 partner as fixed-precision evidence, with the weight in (A2). This is distinct from how an actual level-1 agent processes the partner's speech through Equation (5). The partner's cost-only policy is obtained from (6), (7) and (10) with $\beta = 0$, its current public position and its current reassurance state.

For candidate planning, the proposed factual message and the possible immediate verification outcome are first added to these known-history counts. The predicted partner then responds. Nothing after that response enters the planning window. This order makes the information term and the candidate likelihood refer to the same possible observations.

A2 Bayesian updates and information gain

For latent ℓ and a newly observed variable o , the update minimises

$$F[Q] = \sum_{\ell} Q(\ell) \log \frac{Q(\ell)}{J_i(\ell) P(o | \ell)} = D_{KL}(Q \| P(\ell | o)) - \log P(o). \quad (\text{A3})$$

Thus the minimiser is the exact posterior under the agent's subjective model. This identity does not establish that the subjective model is correct. For a delivered verification, equal likelihood normalisers across the two factual states yield an actual marginal log-odds increment of $+\text{logit } \rho_{\omega_i}$ or its negative; absence of a signal yields zero increment after moves are known.

Writing $o = (E, Y_j)$, the independently checked information calculation is

$$IG_i(y) = \sum_{\ell, o} J_i(\ell) p_i(o | \ell, y) \log \frac{p_i(o | \ell, y)}{\sum_{\ell'} J_i(\ell') p_i(o | \ell', y)}. \quad (\text{A4})$$

It is equivalent to $H(o | y) - E_j H(o | \ell, y)$. Information is computed under the same likelihood used to predict the outcomes; no separate posterior shrink is applied after the expectation. Expected loss can be represented by preferences proportional to the exponential of negative state-action loss, with a normalising constant over the joint domain. This representation does not make the explicit expected loss a separately estimated KL-risk term.

A3 A bound on information from finite private records

After conditioning on the known shared-evidence history, the remaining speech likelihood given x is a non-negative mixture over the partner's fixed private count and profile. Since the profile prior is independent of x , its likelihood ratio between $x=1$ and $x=0$ lies between the smallest and largest private-record likelihood ratios. For N binary records of reliability $\rho_p > 1/2$,

$$|\log BF_{speech}| \leq N \text{logit } \rho_p. \quad (\text{A5})$$

With $N=2$ and $\rho_p=0.75$, the bound is 2.197225. The tested maximum is 2.197222. The bound concerns cumulative speech evidence after subtracting shared-verification updates, not total belief change. Common verification can continue to supply new evidence.

Appendix B Reproducibility and provenance

B1 Annotation re-scoring

The package includes the archived combined engine predictions, direct-prompt predictions, direct-prompt text and a scoring script that produces the confusion counts and Table 2 metrics. Stored runs, gold labels and case identifiers are preserved. The baseline artifact identifies Claude Haiku 4.5; the combined engine artifact has no provider field, although the accompanying forensic note attributes that run to the same model family. Sampling temperature and complete prompt/version hashes are not available. This is an archival comparison, not a fully controlled fresh model evaluation.

Historical non-SHIFT accuracy of about 87% refers to a May build and different aggregation. It is not the July majority score of 127/149 (85.2%). The six-rater pilot values come from the supplied internal pilot report. General methodological references cannot substitute for its unavailable raw rater matrices. Corpus-construction descriptors in Section 2.2 are retained with their documentation status rather than presented as newly verified case-level results.

B2 Dynamics configuration

The main task uses 8 rounds, 2 initial private records per agent, private and verification reliability 0.75, initial prior odds 3:1 in opposite directions, fixed-message reliability 0.70 inside the approximate partner model, $k=2$, $b=1.1$, $\delta=0.12$, $\tau=0.20$, and arrival values 0.05, 0.35 and 0.90. Table 3 gives focal profiles; the internal eight-point dictionary is specified in Section 3.4. Main dynamics use seed 2026093081; the observer worlds use 2026093082, validation 2026093083, and label generation 2026093084.

Effective REASSURE, the local inactive-message control and forced inviting SILENCE occur at zero-based event 7, B’s fourth move. Earlier spontaneous reassurance is unchanged. Permanent removal of c or γ from that event changes actual agent parameters while their internal partner dictionaries remain fixed; these are controlled simulator manipulations, not inferred treatment responses. Forced invitation changes form while retaining the chosen coarse act. The no-gating condition sets arrival to its base rate independently of form.

B3 Observer likelihood and calibration uncertainty

For private counts, the external prior uses

$$P(s_A, s_B) = 1/2 \sum_{x=0}^1 P(s_A | x) P(s_B | x), \quad (B1)$$

then multiplies by the independent uniform profile priors. Within a latent-form path, the sequential predictive probability for a label is

$$P(z_t = z | f_t = f, h_t) = \frac{n_{fz} + \alpha_0 + N_{fz}(h_t)}{\sum_{z'} (n_{fz'} + \alpha_0) + N_f(h_t)}. \quad (B2)$$

Here n_{fz} are historical calibration counts; N_{fz} are labels previously assigned to form f along that proposed path. Counts include both speakers. This integrates a fixed uncertain row over a sequence and is different from independently using a posterior-mean row at every turn. The physical fact is analytically marginalised given both private counts and shared verification. Agent beliefs are reconstructed within each stratum and form history; they are not supplied as observations.

Main forecast losses exclude the first four moves and use natural logarithms. Probabilities are clipped to $[10^{-12}, 1 - 10^{-12}]$ for binary log loss; observed-act probabilities are clipped below at 10^{-12} . Contrasts are paired by world, using dialogue-level average losses. The external A-profile result is based on 24 examples of each generating A point with reference B, rather than a balanced sixteen-pair evaluation.

Appendix C Numerical checks and sensitivity

Table 7. Validation scope. Passing a numerical check establishes an implementation property, not a psychological interpretation.

Check	Recorded result
Independent scalar choice calculation	Maximum probability difference 2.61×10^{-15}
Normalisation and mutual-information identity	Maximum MI difference 2.00×10^{-15}
Bayesian averaging identity	Error 2.08×10^{-17}
Reliability parameterisation	Pass
Role swap with reversed order	Maximum difference 1.44×10^{-15}
No repeated creation of private records	Speech log Bayes factor remains within (A5)
Current public position is causal	Changing it changes the policy
Local inactive-message timing	No pre-intervention history difference
Form-to-access coupling removed	Conditional form difference 2.22×10^{-16}
Sequential Dirichlet integration	Identity error zero
Particle filter versus 576 exact short paths	Posterior error 0.00221; forecast error 0.00281
Uninformative labels	Null-channel check passes
Known-form arrival oracle	Arrival-rate error zero

Particle sensitivity uses sixteen dialogues and $K \in \{4, 8, 32\}$, retaining all static strata. Increasing K from 8 to 32 changes mean arrival log loss by -0.00658 for acts plus evidence and -0.01273 for labels. These are local sensitivity results, not a convergence certificate for all 96 dialogues.

Calibration-prior sensitivity evaluates channel information under an equal form prior using pseudocounts 0.1, 0.5 and 1.0. The median sampled channel mutual information is respectively 0.669, 0.514 and 0.384 nats. This checks sensitivity of the assumed measurement channel; it does not rerun the full external observer under all three priors. The four sensor-reliability runs and all 49 primary dynamics conditions are retained in the supplied results. The package records the tested software environment and file hashes so the specification, figures and tables can be traced to one implementation.